

Electricity Demand Forecasting using Deep Learning

Project Overview

The objective of this project is to build an end-to-end electricity demand forecasting system. The workflow begins with understanding the time-series data and continues through preprocessing, exploratory data analysis, feature engineering, deep learning, validation, evaluation, and dashboard development.

Phase 1 – Data Understanding

The first phase focuses on understanding the structure and behavior of the electricity demand data. The timestamp column is converted to datetime format, records are sorted chronologically, and the time interval between observations is checked.

- Analyze timestamps and hourly time intervals.
- Check missing and duplicate records.
- Analyze long-term electricity demand trends.
- Identify daily, weekly, monthly, and annual patterns.
- Identify abnormal or unusually high demand periods.

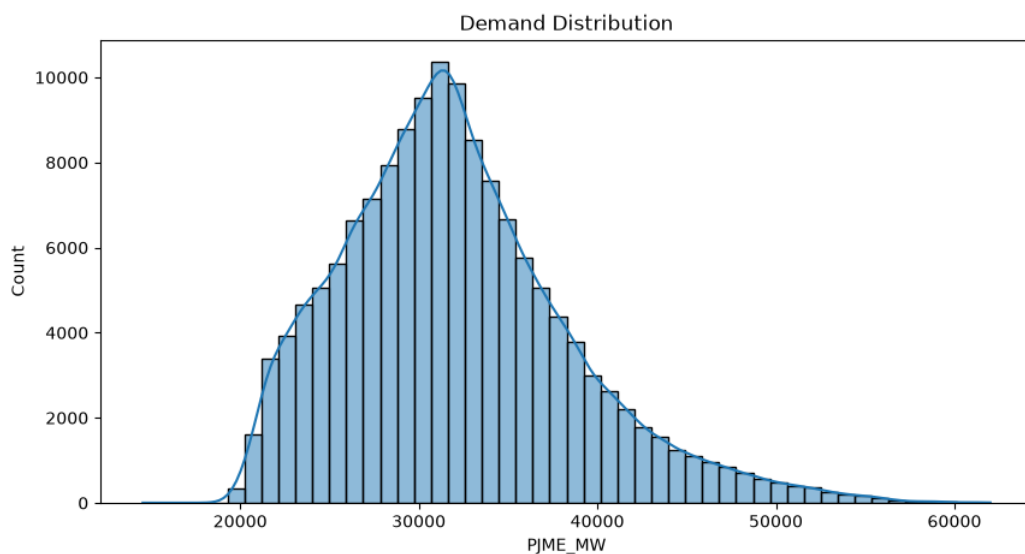


Figure 1: Demand Distribution

Phase 2 – Time-Series Preprocessing

The preprocessing phase prepares the raw time-series for machine learning. The data is sorted chronologically and missing timestamps or missing demand values are checked before modelling.

- Chronologically sort the data.
- Check and handle missing timestamps.
- Handle missing demand values.
- Detect and treat abnormal values where appropriate.
- Create time-based features such as hour, day, week, month, and day of week.
- Create lag and rolling statistical features.

Rolling features capture recent demand behavior. A 24-hour rolling mean represents the recent daily demand level, while a 168-hour rolling mean represents the recent weekly demand level.

Phase 3 – Exploratory Data Analysis

EDA is used to understand demand patterns before model development. Hourly profiles, weekly seasonality, rolling statistics, demand distributions, and autocorrelation can reveal important time-series relationships.

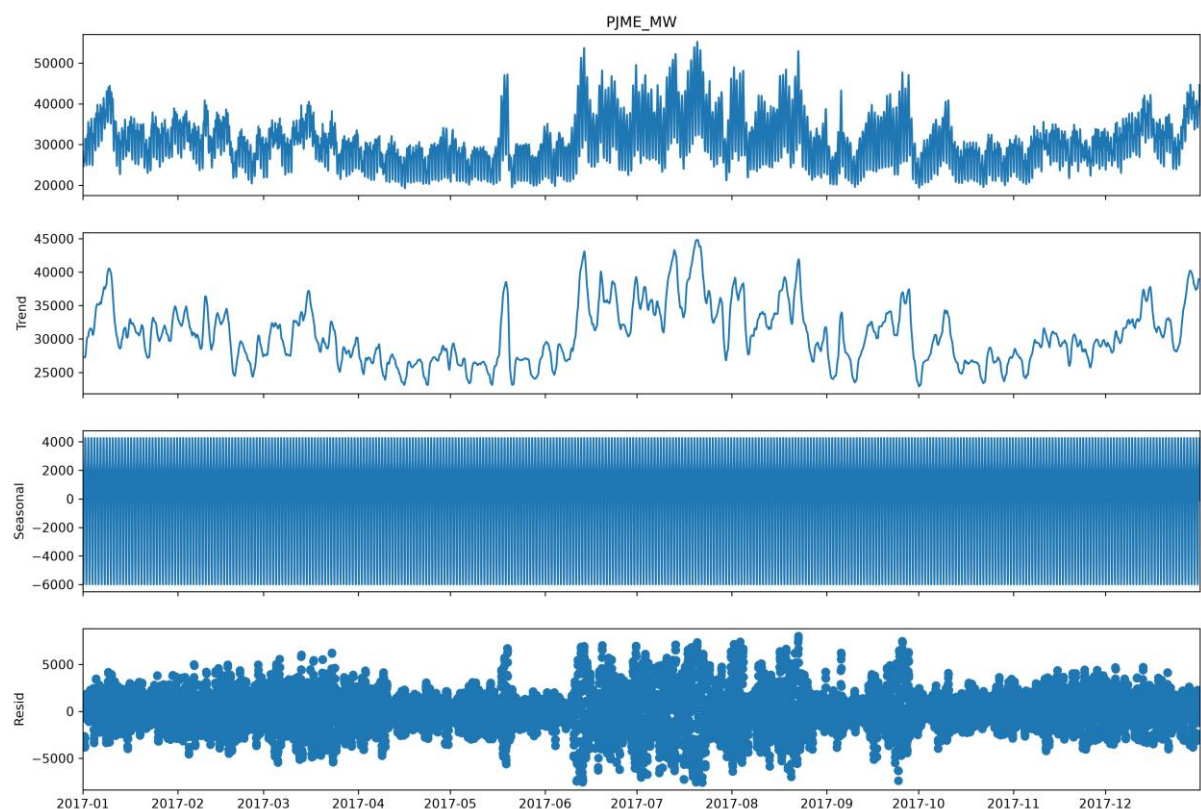


Figure 2: Seasonal Decomposition

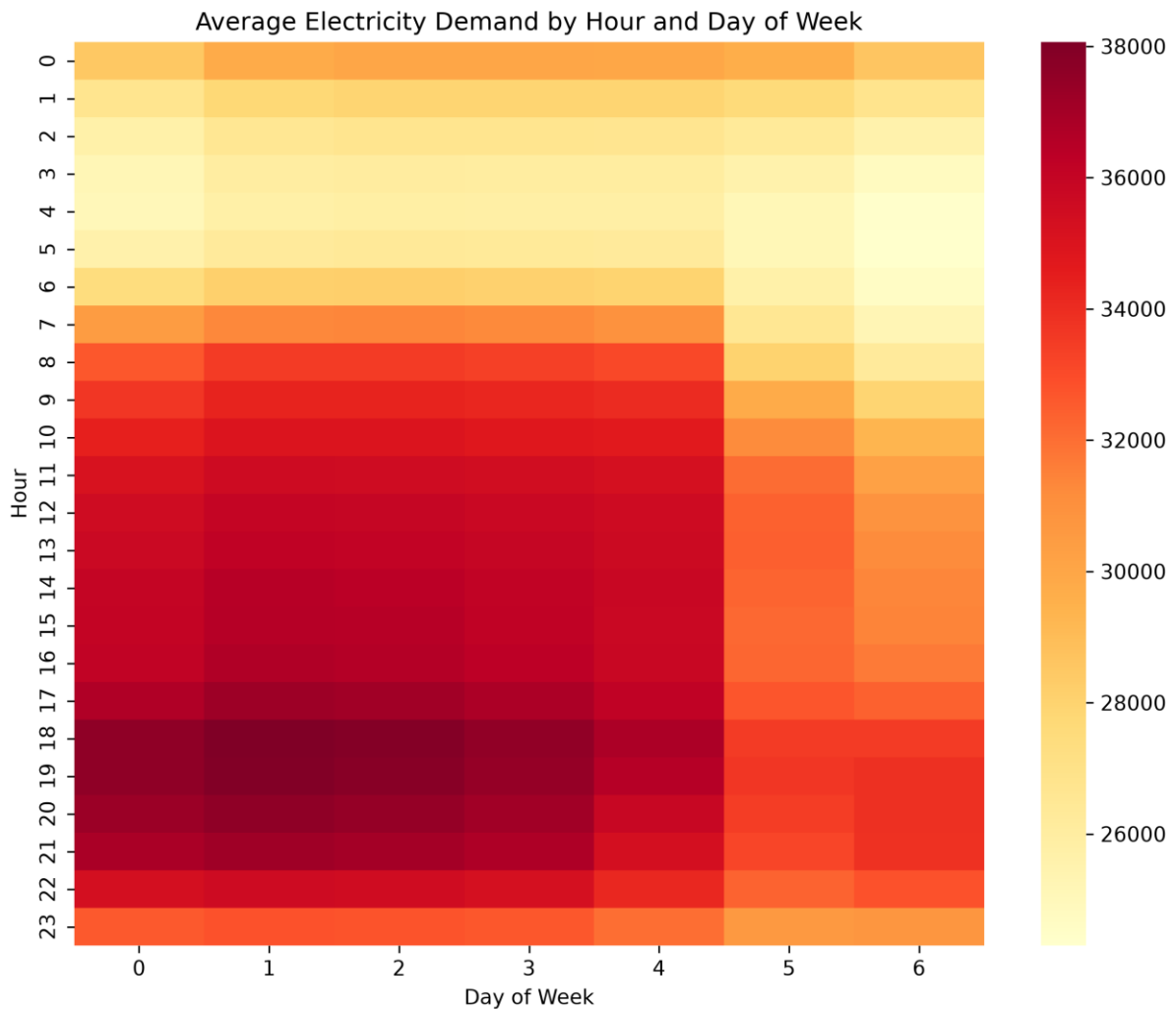


Figure 3: hour day heatmap

Phase 4 – Baseline Forecasting Models

Before applying deep learning, simple forecasting methods are created as baseline models. These provide a reference point for determining whether the deep learning models provide meaningful improvement.

Naive Forecast: Uses the most recent observed demand as the forecast.

Moving Average: Uses the average of recent demand observations.

Previous-Day: Uses demand from the corresponding previous day.

Previous-Week: Uses demand from the corresponding previous week.

No.	Model	MAE	RMSE	MAPE (%)	R ²
1	Naive	2206.33	3027.41	7.01	0.779
2	Moving Average	3558.37	4468.46	11.72	0.519
3	Previous Day	2206.33	3027.41	7.01	0.779
4	Previous Week	3434.42	4735.14	10.70	0.460

Phase 5 – Deep Learning Models

The deep learning phase uses historical sequences to predict future electricity demand. For example, the model can use the previous 24, 48, or 168 hours to predict the next 24 hours.

Models :

- Simple RNN - learns sequential relationships using recurrent hidden states.
- LSTM - uses gates and memory cells to retain useful long-term information.
- GRU - uses a simpler gated recurrent architecture with fewer gates than LSTM.
- Bidirectional LSTM - processes the sequence in both directions within each input sequence.

Model	Input Hours	Output Hours	MAE	RMSE	MAPE (%)	R ²	Bias
RNN	24	24	1376.72	1926.89	4.35	0.911	78.36
GRU	48	24	1408.35	2007.91	4.42	0.903	130.56
RNN	48	24	1427.28	1991.72	4.54	0.904	291.56
GRU	24	24	1429.70	2086.47	4.50	0.895	241.55
RNN	168	24	1430.21	1981.73	4.58	0.905	375.38
GRU	168	24	1452.68	2075.13	4.60	0.896	284.29
LSTM	48	24	1480.84	2166.63	4.64	0.887	244.60
LSTM	24	24	1491.63	2165.56	4.69	0.887	315.43
Bi-LSTM	24	24	1532.42	2220.76	4.83	0.881	225.46
LSTM	168	24	1539.11	2171.03	4.91	0.886	447.83
Bi-LSTM	168	24	1557.67	2229.98	4.93	0.880	369.86

Selected Model: RNN (24-hour input → 24-hour output)

The RNN with 24-hour input and 24-hour output is selected as the best-performing deep learning model because it achieved the lowest MAE, RMSE, and MAPE, along with the highest R² (0.911) among the evaluated deep learning models.

Phase 6 – Model Improvement

After establishing baseline deep learning models, additional experiments are performed to improve forecasting performance.

- Compare different sequence lengths such as 24, 48, and 168 hours.
- Add lag features and rolling statistical features.
- Scale input features using an appropriate scaler.
- Experiment with dropout and batch normalization.
- Compare optimizers such as Adam and RMSprop.
- Use learning-rate scheduling.
- Use early stopping when appropriate.
- Tune batch size, number of layers, and number of neurons.
- Perform hyperparameter tuning where required.

Optimized Deep Learning Model Comparison

Architecture	Model	Input Hours	Output Hours	MAE	RMSE	MAPE (%)	R ²	Bias
RNN + RMSprop + Learning Rate	RNN	168	24	1391.47	1955.14	4.43	0.9078	260.425411
RNN Baseline	RNN	24	24	1376.72	1926.89	4.35	0.9105	78.356911
RNN + RMSprop	RNN	48	24	1368.04	1924.23	4.32	0.9108	30.372446
LSTM 32 Neurons	LSTM	48	24	1400.05	1995.07	4.41	0.9041	89.953964
LSTM + EarlyStopping	LSTM	24	24	1371.59	1966.05	4.31	0.9068	-7.358136
GRU 32 Neurons	GRU	168	24	1393.96	1965.06	4.40	0.9069	133.021532
GRU + EarlyStopping	GRU	48	24	1378.18	1931.76	4.37	0.9101	178.304365
GRU + EarlyStopping	GRU	24	24	1375.32	1951.27	4.31	0.9082	-55.753064
Bi-LSTM + EarlyStopping	Bi-LSTM	48	24	1396.55	1976.80	4.40	0.9058	98.702275

Selected Best Model:

RNN + RMSprop (48 Hours → 24 Hours) was selected as the best-performing model after comparing all Phase 6 models across MAE, RMSE, MAPE, R², and Bias. It achieved the lowest MAE (1368.04 MW), lowest RMSE (1924.23 MW), lowest MAPE (4.3165%), highest R² (0.910843), and a very low Bias (30.37 MW), demonstrating the best overall forecasting performance among the evaluated models.

Phase 7 – Forecast Validation

The selected RNN + RMSprop model (48-hour input → 24-hour output) was validated using a rolling forecast approach over 30-day and 60-day periods.

period	MAE	RMSE	MAPE	R2	Bias
30 days	1760.444645	2485.47334	5.014282	0.899805	304.760352
60 days	1945.609527	2739.147156	5.341999	0.873378	282.692963

Summary:

The model achieved 5.01% MAPE and 0.900 R^2 for 30 days and 5.34% MAPE and 0.873 R^2 for 60 days, demonstrating good forecasting performance on unseen data.

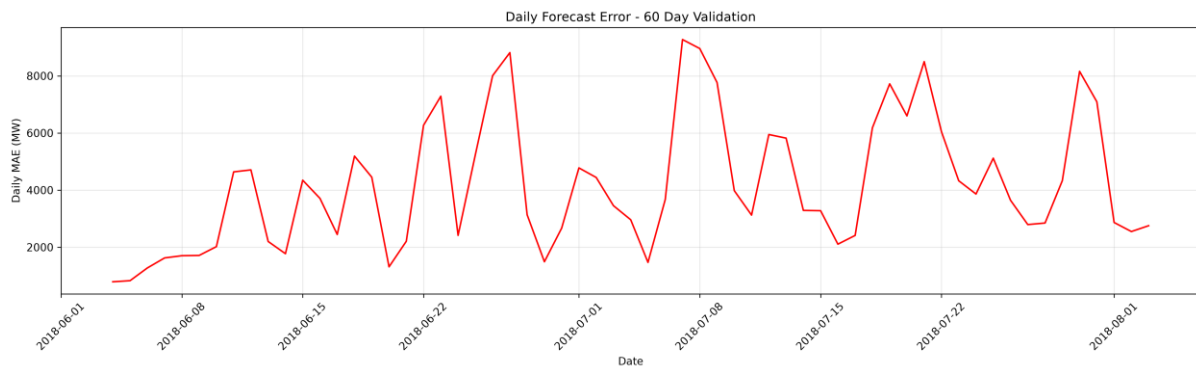


Figure 4: daily error

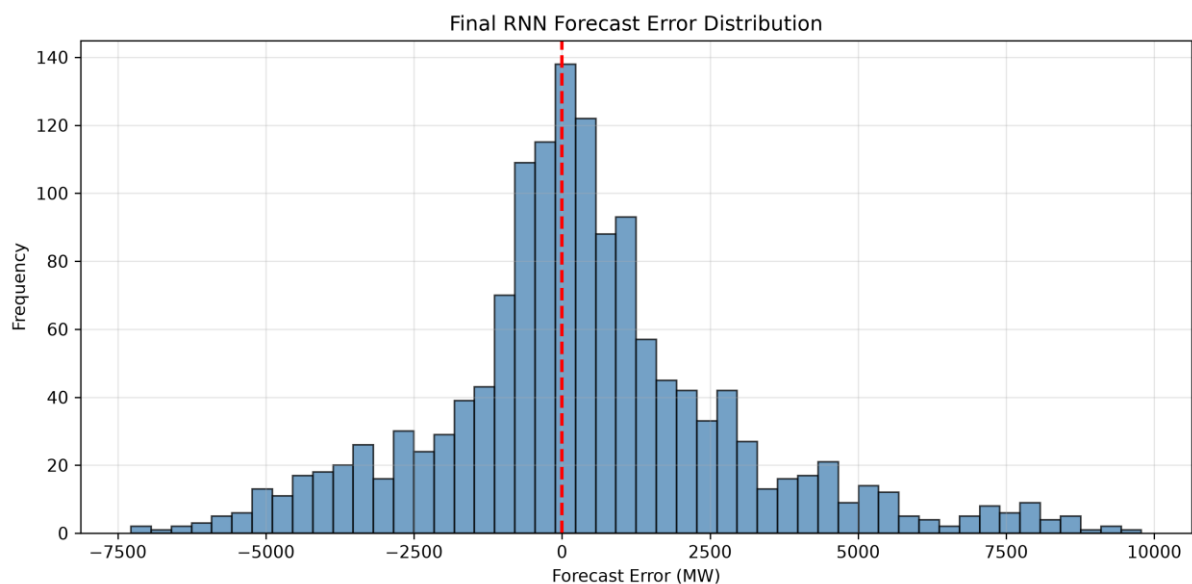


Figure 5: Forecast Error Distribution

Phase 8 – Model Evaluation

The models are evaluated using MAE, MSE, RMSE, MAPE, R^2 , and forecast bias. Lower MAE, RMSE, and MAPE are preferred, while higher R^2 is preferred. Bias indicates whether the model systematically overpredicts or underpredicts demand.

Metrics	Purpose
MAE	Average absolute forecasting error.
MSE	Average squared forecasting error.
RMSE	Square root of MSE; emphasizes larger errors.
MAPE	Average percentage forecasting error.
R^2	Measures explained variation.
Bias	Shows systematic overprediction or underprediction.

Best Model Evaluation

Architecture	Model	Input Hours	Output Hours	MAE	RMSE	MAPE (%)	R^2
RNN + RMSprop	RNN	48	24	1368.04	1924.23	4.32	0.9108

Best Model: RNN + RMSprop (48-hour input → 24-hour output) — selected based on its lowest MAE, RMSE, and MAPE with an R^2 of 0.9108.

Phase 9 – Forecasting Dashboard

The Flask web application provides an interactive platform for **electricity demand forecasting**, model performance comparison, validation analysis, and exploration of historical electricity consumption using the selected **RNN + RMSprop model**.

Dashboard

- Latest Electricity Demand
- Historical Consumption Overview
- Average Demand
- Maximum Demand
- Minimum Demand
- Forecast Summary
- Best Model Information

Forecasting Module

- Forecast Horizon Selection
- **48-Hour Historical Input Sequence**
- **24-Hour Demand Forecast**
- Forecasted Demand Values
- Peak Forecast Analysis
- Minimum Forecast Analysis
- **95% Forecast Intervals**
- Forecast Visualization

Analytics

- Historical Demand Trends
- Hourly Demand Analysis
- Daily Demand Trends
- Weekly/Monthly Demand Patterns
- Rolling Statistics
- Peak Demand Analysis
- Forecast Analytics
- Model Performance Insights

Validation

- 30-Day Forecast Validation
- 60-Day Forecast Validation
- MAE, RMSE and MAPE
- R^2 and Forecast Bias
- Peak Demand Error
- Forecast Error Analysis
- Validation Performance Comparison
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Technologies Used

- Python
- Flask
- Pandas
- NumPy
- TensorFlow / Keras
- Scikit-learn
- Matplotlib
- Seaborn
- HTML5

- CSS3
- Bootstrap 5
- Chart.js
- Git
- GitHub

Application Workflow

Historical Electricity Demand



Time-Series Preprocessing



Feature Engineering



14 Input Features



48-Hour Historical Sequence



Feature Scaling



RNN + RMSprop Model



24-Hour Demand Forecast



Inverse Scaling



Forecast Intervals



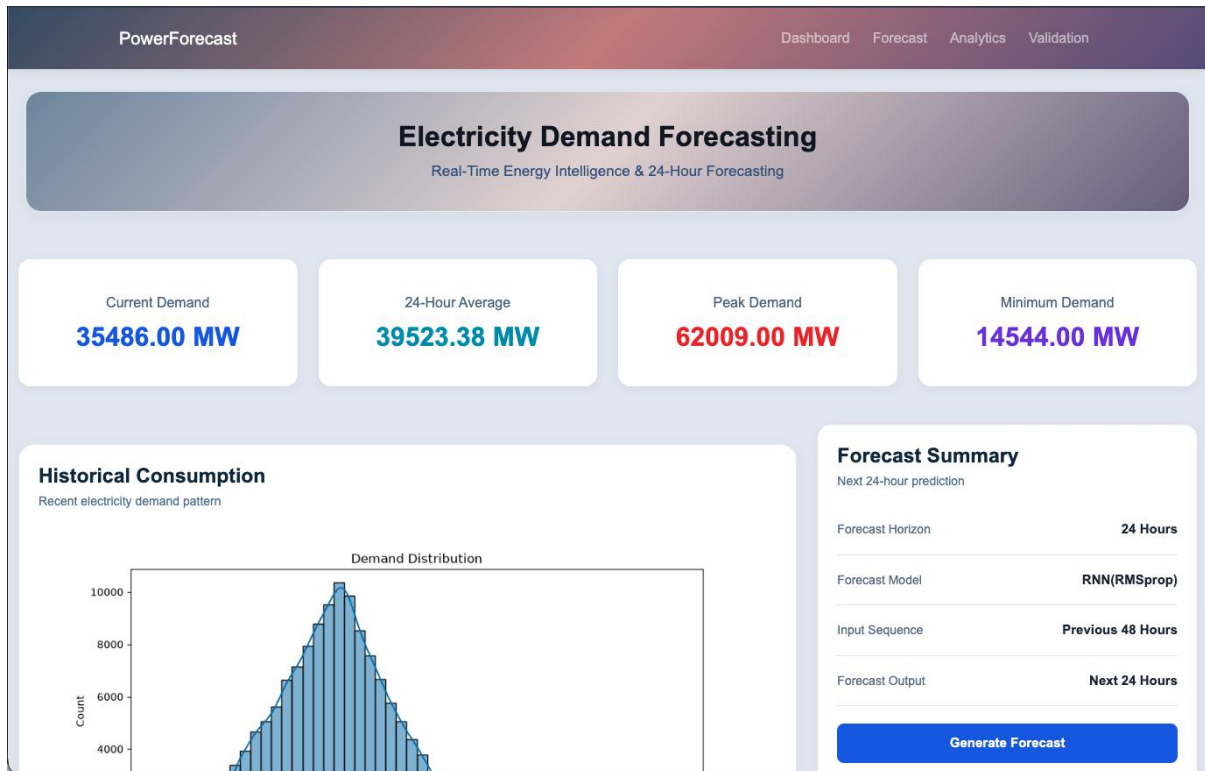
Display Forecast Results



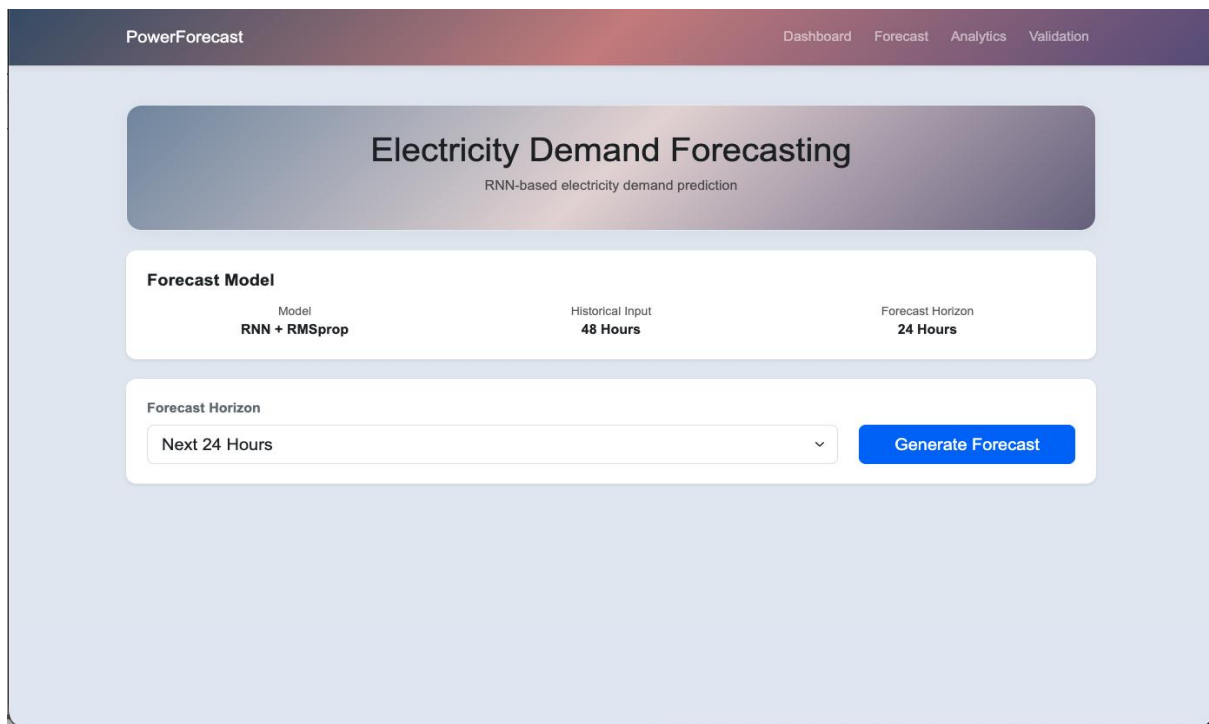
Analytics & Validation

Screenshots:

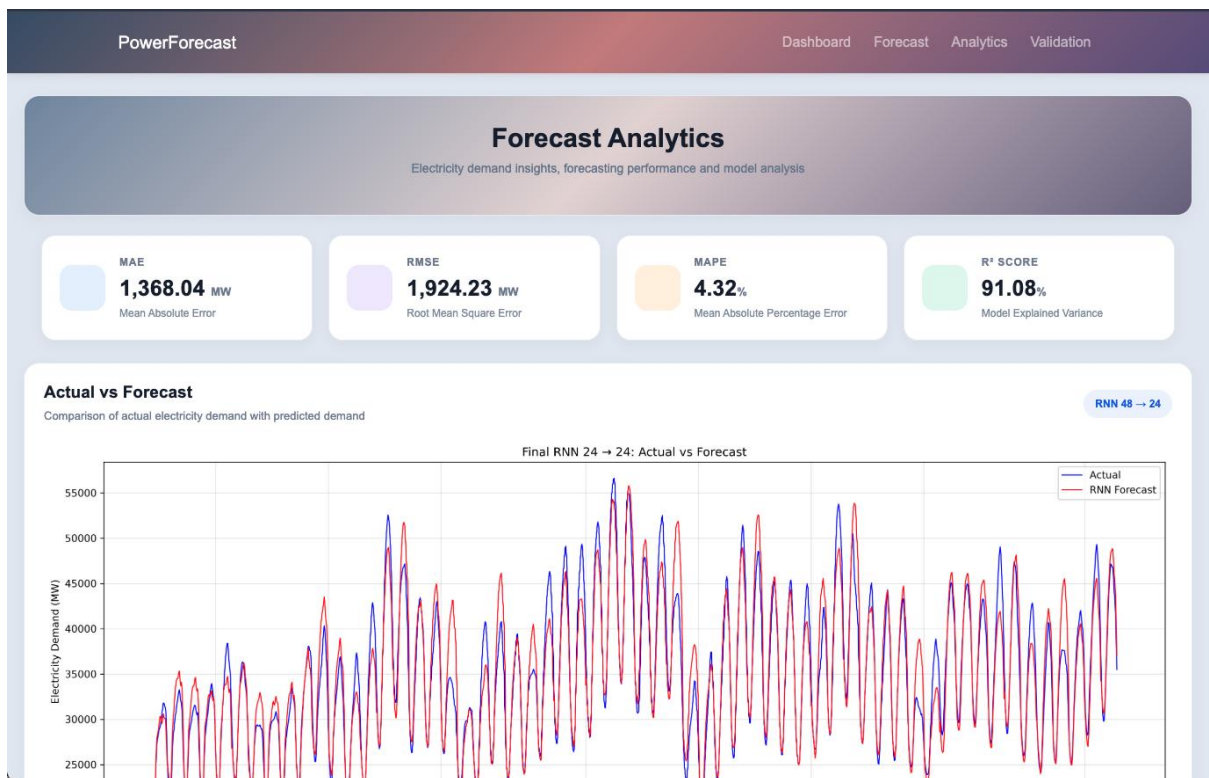
Dashboard



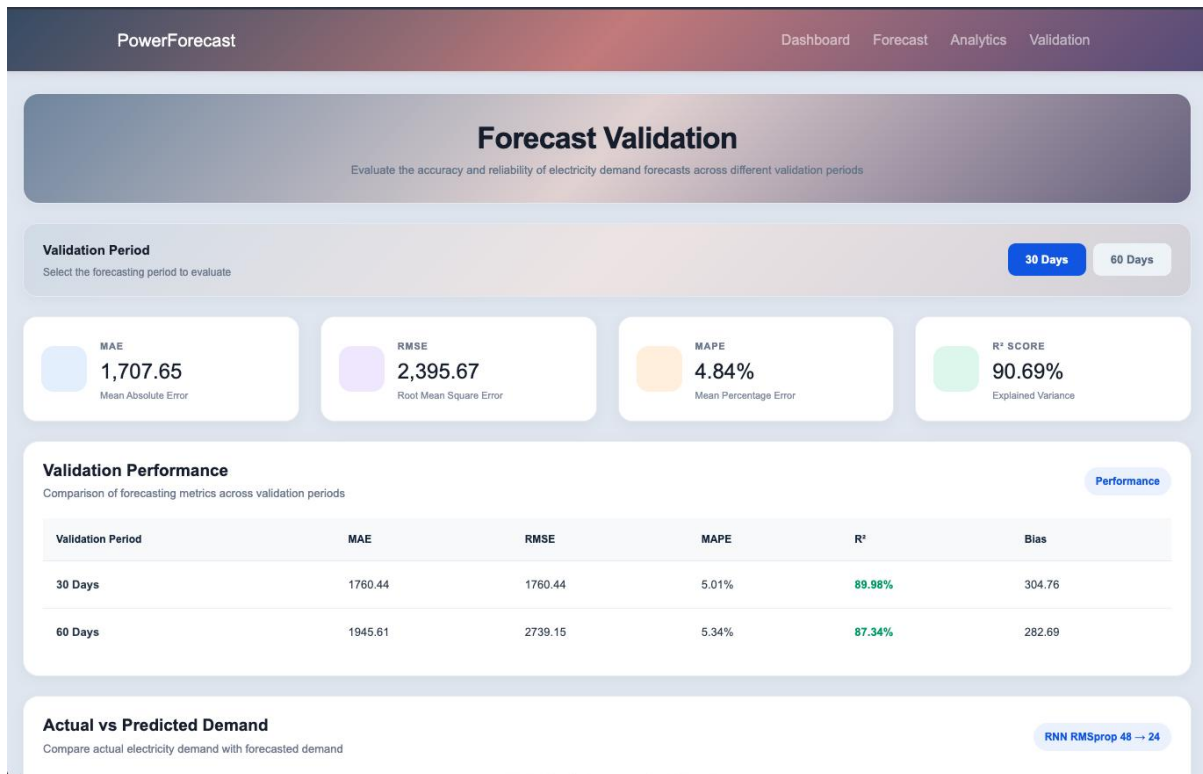
Forecast



Analytics



Validation



Conclusion

This project demonstrates a complete end-to-end electricity demand forecasting workflow. The process starts with time-series data understanding and preprocessing, continues through exploratory analysis and feature engineering, and then applies baseline and recurrent deep learning models.

RNN, LSTM, GRU, and Bidirectional LSTM models are compared using multiple forecasting metrics. Model improvement experiments investigate sequence length, rolling features, scaling, optimization, learning-rate scheduling, regularization, and hyperparameter tuning.